

# Andrew W. Shenstone

Shenstone, Andrew - Application for Staff Software Engineer, Front-End @ Dutchie

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## SUMMARY

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I am applying for Staff Software Engineer, Front-End on Dutchie's Ecommerce Platform because the work maps directly to Bloom: a regulated retail category where operators (wineries) and consumers both depend on software that cannot break compliance, checkout, or trust. I built the greenfield **Next.js** admin, shipped against **Ruby on Rails GraphQL** shared across web and mobile, and lived inside **Shopify Plus** integrations and Polaris-aligned UI with design—retailer success driving platform success, the same two-sided story Dutchie describes for dispensaries.

I have 12+ years in software engineering. **React** has been my primary UI stack from agency work through Bloom and founder delivery; **TypeScript** is what I ship in production today. I also have roughly **six years on Ruby on Rails** (through **Rails 6**) and am comfortable in a **Rails** console and across the monolith—not only the frontend—while catching up on changes since Rails 6. I lead as a hands-on engineer through mentoring, pairing, and review (no direct reports). **Apollo Client** and **Styled Components** are house-tool ramps; cannabis labeling is new, but state rules, operator tooling, and ordering UX are not.

This team is the closest match in my background to what you are building now—regulated retail ecommerce on a long-lived platform, not a surface-level stack overlap.

## SKILLS

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**Ecommerce & platform (Bloom transfer):** Regulated retail ecommerce: alcohol commerce at Bloom (Shopify Plus, operator + consumer flows), React and TypeScript for mission-critical ordering and admin experiences, Ruby on Rails (~6 years through Rails 6): productive in the console and across the app; recent work is React-heavy but I jump into Rails codebases fast and ramp on newer Rails versions, React Native

production apps (Ply)—relevant to multi-surface ecommerce (web, mobile, kiosk/SDK contexts)

**GraphQL, data & quality:** GraphQL in production (Bloom shared API; Tilia platform API—schema through typed clients), Apollo Client: ramp on house client; fundamentals already in daily practice, MongoDB: 1+ year production on Plantfolio before late migration to Postgres—know it well; aligns with Dutchie’s Mongo + Postgres mix, PostgreSQL and Prisma on Neon today (Plantfolio); SQL reporting at Bloom, Vitest, React Testing Library, Cypress—clean, tested, organized code

**Staff-level delivery:** Mentoring through pairing, code review, and standards (Chakra adoption lead at Tilia; informal tech lead at Bloom—no direct reports), Cross-functional scoping with product and design when the ideal path is not available, Component-driven UI; Styled Components adoptable (recent default: Tailwind/Chakra)

**AI/ML:** OpenAI API, AI SDK, Vector Embeddings, Natural Language Processing, Retrieval Augmented Generation (RAG)

## EXPERIENCE

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### Founder & Lead Engineer

Plantfolio | November 2023 - Present

#### Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

#### Technical Achievements

- **AI Integration:** Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- **Modern Frontend Architecture:** Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- **Data modeling & migration:** Built Plantfolio on MongoDB and shipped production there for 1+ year, then migrated late to PostgreSQL on Neon (15+ interconnected models with Prisma today)—deep familiarity with document and relational modeling tradeoffs
- **Database choices:** Evaluated Supabase during exploration but did not ship on it; chose Neon for managed Postgres; auth and subscription tiers via Clerk with Postgres-centric access patterns in production today
- **Content Management:** Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

## **Scale & Performance**

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

## **Client Case Study - AI Signage**

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup

- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

## **Full-Stack Ownership**

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

## **Code Quality & Leadership**

- Established coding standards and architectural patterns that enabled rapid, consistent development
  - Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
  - Built reusable component library and design system, ensuring consistent UI/UX across the platform
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## **Senior Software Engineer**

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
- Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
- Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns

- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
  - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
  - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
  - Role ended in a company-wide layoff affecting approximately one-third of employees
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## Senior Software Engineer

Bloom.wine | September 2020 - October 2022

### Ecommerce platform (regulated retail)

- Architected the greenfield Next.js admin for Bloom's primary operator product—retailer-facing tooling where wineries ran clubs, inventory, and compliance-sensitive workflows; much of that stack remains in production across many clients
  - Worked across the Rails stack (~6 years through Rails 6, Bloom and earlier agency): GraphQL APIs, models, and the console—not only the React layer; can jump into both sides quickly and ramp on post-Rails 6 conventions
  - Platform shape Dutchie would recognize: owned experiences, third-party surfaces, and API-driven flows consumed by web and mobile
  - All winery clients on Shopify Plus—integrations (Shopify, Stripe, Shipstation, Mailchimp, ActiveCampaign) and Polaris-aligned UI with design; alcohol is a heavily regulated category with state-by-state constraints analogous to cannabis retail complexity
  - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports); collaborated with product and design in Figma on component standards
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## Software Engineer II

## **Tune Inc - Marketplace Automation** | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
  - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
  - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
  - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
  - Role ended in a company-wide layoff
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## **Software Developer**

**Ply Interactive Inc** | June 2014 - July 2019

### **React Native**

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

### **Web & platforms**

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects

- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

## COMPANY REFERENCE

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**Plantfolio** (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - [plantfolio.garden/](https://plantfolio.garden/)

**Tilia** (2023) - All-in-one payment platform for digital economies - [tilia.io/](https://tilia.io/)

**Bloom.wine** (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - [bloom.wine/](https://bloom.wine/)

**Tune Inc - Marketplace Automation** (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - [tune.com/](https://tune.com/)

**Ply Interactive Inc** (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)